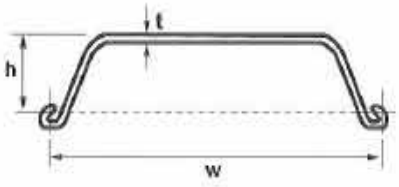
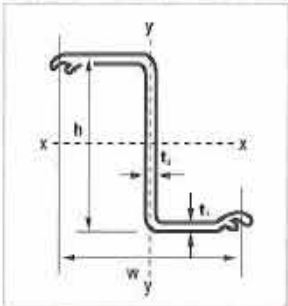
Dimensions and Properties

i) U Type



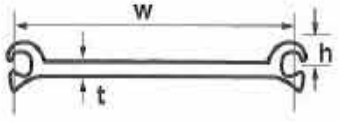
Type	Dimensions			Sectional Area	Weight		Moment Of Inertia		Section Modulus	
	Width w mm	Height h mm	Thickness t mm	Per Pile cm^2	Per Pile kg / m	Per Linear Meter Of Wall kg / m^2	Per Pile cm^4	Per Linear Meter Of Wall cm^4 / m	Per Pile cm^3	Per Linear Meter Of Wall cm^3 / m
I A	400	85	8.0	45.2	35.5	88.8	598	4,500	88	529
II	400	100	10.5	61.2	48.0	120	1,190	8,700	146	870
II A	400	120	9.2	55.0	43.2	108	1,450	10,600	162	880
III	400	130	13.0	76.4	60.0	150	2,320	17,400	232	1,340
III A	400	150	13.1	74.4	58.4	146	2,840	22,800	253	1,520
IV	400	160	16.0	96.9	76.1	190	4,110	34,400	334	2,150
IV A	400	180	16.2	94.2	74.0	185	4,910	40,600	374	2,250
V L	500	200	24.3	133.8	105.0	210	7,960	63,000	520	3,150
VI L	500	225	27.6	153.0	120.0	240	11,400	86,000	680	3,820

ii) Z Type



Type	Dimensions				Sectional Area	Weight		Moment Of Inertia		Section Modulus	
	Width w mm	Height h mm	Thickness t_1 mm	Thickness t_2 mm	Per Pile cm^2	Per Pile kg / m	Per Linear Meter Of Wall kg / m^2	Per Pile cm^4	Per Linear Meter Of Wall cm^4 / m	Per Pile cm^3	Per Linear Meter Of Wall cm^3 / m
Z 25	400	305	13.0	9.6	94.32	74.0	185	15,300	38,300	1,000	2,510
Z 32	400	344	14.2	10.4	107.7	84.5	211	21,900	54,900	1,280	3,190
Z 38	400	364	17.2	11.4	122.2	96.0	240	27,700	69,200	1,520	3,800
Z 45	400	367	21.9	13.2	148.2	116.0	290	33,400	83,500	1,820	4,550

iii) Flat Type



Type	Dimensions			Sectional Area	Weight		Moment Of Inertia		Section Modulus	
	Width w mm	Height h mm	Thickness t mm	Per Pile cm^2	Per Pile kg / m	Per Linear Meter Of Wall kg / m^2	Per Pile cm^4	Per Linear Meter Of Wall cm^4 / m	Per Pile cm^3	Per Linear Meter Of Wall cm^3 / m
F	400	44.5	9.5	69.07	54.2	136	190	525	47.8	120
F A	400	44.5	12.7	77.50	60.8	152	196	520	48.3	117

Standard Specifications

Mechanical Properties

(Grade SY295 or SY390) **JIS A 5528:2000**

Type of Product	Classifications		Mechanical Properties		
			Yield Point N/mm² (min.)	Tensile Strength N/mm²	Elongation % (min.)
Sheet Pile	JIS A 5528 : 2000	SY295	295	490 min	17
		SY390	390	540 min	15

Chemical Compositions

Type of Product	Classifications		Chemical Compositions (ladle analysis), %									
			C (max.)	Si (max.)	Mn (max.)	P (max.)	S (max.)	CE[E] (max.)	SW[F] (max.)	Nb (max.)	V (max.)	N (max.)
Sheet Pile	JIS A 5528 : 2000	SY295	-	-	-	0.040	0.040	-	-	-	-	-
		SY390	-	-	-	0.040	0.040	-	-	-	-	-

Tolerances

JIS A 5528 : 2000		
Hot Rolled Steel Sheet Pile "U" Shape		
Dimension		**Tolerance**
Height		± 4%
Width (B)		+ 10 - 5
Thickness	t < 10	± 1.0
	10 ≤ t < 16	± 1.2
	t ≤ 16	± 1.5
Length (L)		+Not Specified 0
Deflection	L ≤ 10 m	Full Length (M) x 0.12% max.
	L > 10 m	Full Length - 10 m x 0.10% + 12 mm max.
Camber	L ≤ 10 m	Full Length (M) x 0.25% max.
	L > 10 m	Full Length - 10 m x 0.20% + 25 mm max.
Difference in Vertically Cut Sections		Within 4% of Width

Note : The deflection shall be in the direction parallel to the sheet pile wall and the camber shall be in the direction vertical to the sheet pile wall.